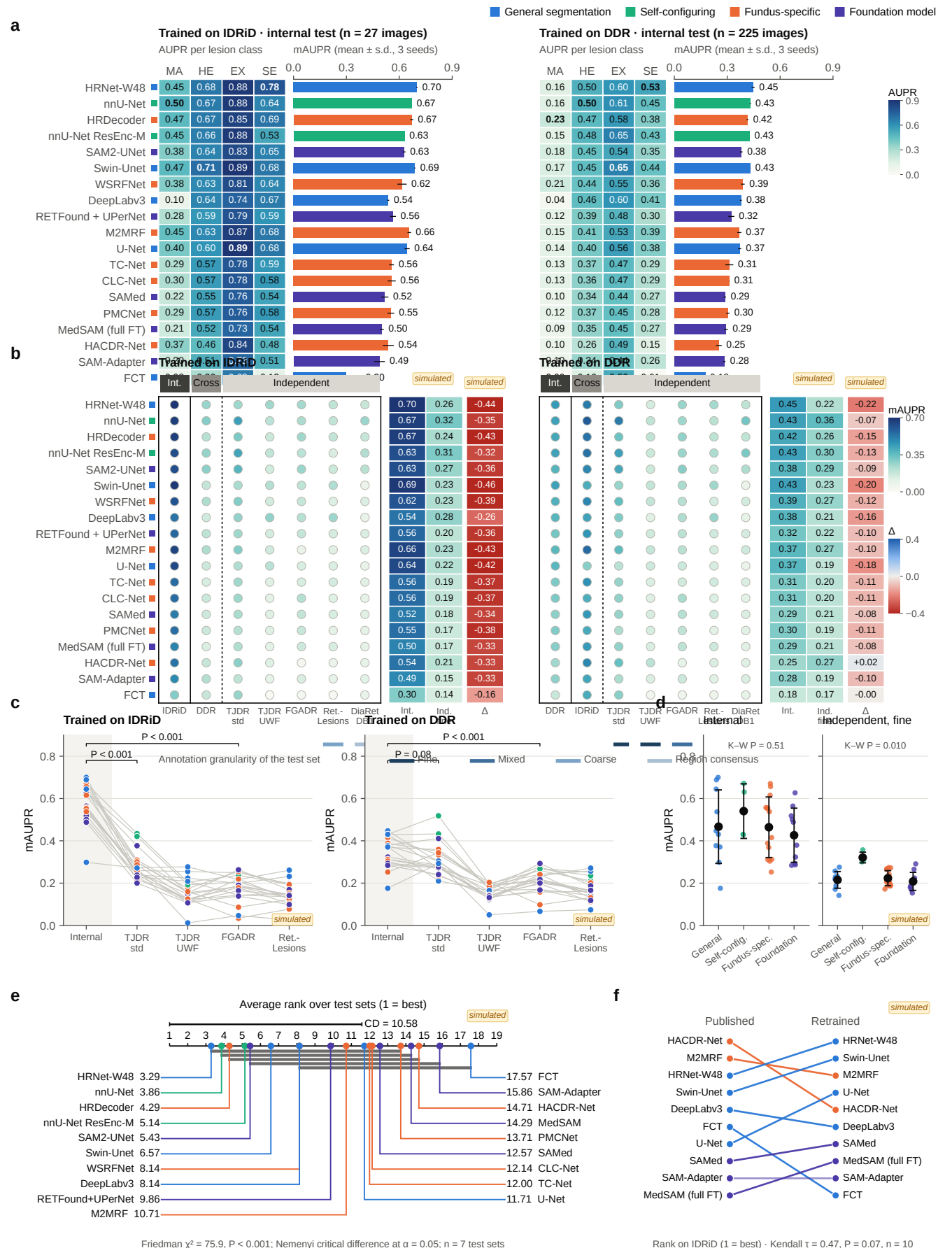

General segmentation
Self-configuring
Fundus-specific
Foundation model
a
Trained on IDRiD · internal test (n = 27 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
HRNet-W48 0.45 0.68 0.88 0.78 0.70
nnU-Net 0.50 0.67 0.88 0.64 0.67
HRDecoder 0.47 0.67 0.85 0.69 0.67
nnU-Net ResEnc-M 0.45 0.66 0.88 0.53 0.63
SAM2-UNet 0.38 0.64 0.83 0.65 0.63
Swin-Unet 0.47 0.71 0.89 0.68 0.69
WSRFNet 0.38 0.63 0.81 0.64 0.62
DeepLabv3 0.10 0.64 0.74 0.67 0.54
RETFound + UPerNet 0.28 0.59 0.79 0.59 0.56
M2MRF 0.45 0.63 0.87 0.68 0.66
U-Net 0.40 0.60 0.89 0.68 0.64
TC-Net 0.29 0.57 0.78 0.59 0.56
CLC-Net 0.30 0.57 0.78 0.58 0.56
SAMed 0.22 0.55 0.76 0.54 0.52
PMCNet 0.29 0.57 0.76 0.58 0.55
MedSAM (full FT) 0.21 0.52 0.73 0.54 0.50
HACDR-Net 0.37 0.46 0.84 0.48 0.54
SAM-Adapter 0.20 0.51 0.51 0.49
FCT
Trained on DDR · internal test (n = 225 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
0.16 0.50 0.60 0.53 0.45
0.16 0.50 0.61 0.45 0.43
0.23 0.47 0.58 0.38 0.42
0.15 0.48 0.65 0.43 0.43
0.18 0.45 0.54 0.35 0.38
0.17 0.45 0.65 0.44 0.43
0.21 0.44 0.55 0.36 0.39
0.04 0.46 0.60 0.41 0.38
0.12 0.39 0.48 0.30 0.32
0.15 0.41 0.53 0.39 0.37
0.14 0.40 0.56 0.38 0.37
0.13 0.37 0.47 0.29 0.31
0.13 0.36 0.47 0.29 0.31
0.10 0.34 0.44 0.27 0.29
0.12 0.37 0.45 0.28 0.30
0.09 0.35 0.45 0.27 0.29
0.10 0.26 0.49 0.15 0.25
0.26 0.28
AUPR
b
Trained on IDRiD
Int. Cross Independent
simulated
IDRiD DDR TJDR std TJDR UWF FGADR Ret.-Lesions DiaRet DB1
Int. Ind. Δ
HRNet-W48 0.70 0.26 -0.44
nnU-Net 0.67 0.32 -0.35
HRDecoder 0.67 0.24 -0.43
nnU-Net ResEnc-M 0.63 0.31 -0.32
SAM2-UNet 0.63 0.27 -0.36
Swin-Unet 0.69 0.23 -0.46
WSRFNet 0.62 0.23 -0.39
DeepLabv3 0.54 0.28 -0.26
RETFound + UPerNet 0.56 0.20 -0.36
M2MRF 0.66 0.23 -0.43
U-Net 0.64 0.22 -0.42
TC-Net 0.56 0.19 -0.37
CLC-Net 0.56 0.19 -0.37
SAMed 0.52 0.18 -0.34
PMCNet 0.55 0.17 -0.38
MedSAM (full FT) 0.50 0.17 -0.33
HACDR-Net 0.54 0.21 -0.33
SAM-Adapter 0.49 0.15 -0.33
FCT 0.30 0.14 -0.16
Trained on DDR
Int. Cross Independent
simulated
DDR IDRiD TJDR std TJDR UWF FGADR Ret.-Lesions DiaRet DB1
Int. Ind. Δ
0.45 0.22 -0.22
0.43 0.36 -0.07
0.42 0.26 -0.15
0.43 0.30 -0.13
0.38 0.29 -0.09
0.43 0.23 -0.20
0.39 0.27 -0.12
0.38 0.21 -0.16
0.32 0.22 -0.10
0.37 0.27 -0.10
0.37 0.19 -0.18
0.31 0.20 -0.11
0.31 0.20 -0.11
0.29 0.21 -0.08
0.30 0.19 -0.11
0.29 0.21 -0.08
0.25 0.27 +0.02
0.28 0.19 -0.10
0.18 0.17 -0.00
mAUPR
Δ
c
Trained on IDRiD
Trained on DDR
P < 0.001
P < 0.001
P = 0.08
Annotation granularity of the test set
Fine
Mixed
Coarse
Region consensus
mAUPR
Internal
TJDR std
TJDR UWF
FGADR
Ret.-Lesions
simulated
d
Internal
Independent, fine
K–W P = 0.51
K–W P = 0.010
mAUPR
General
Self-config.
Fundus-spec.
Foundation
simulated
e
Average rank over test sets (1 = best)
CD = 10.58
simulated
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
HRNet-W48 3.29
nnU-Net 3.86
HRDecoder 4.29
nnU-Net ResEnc-M 5.14
SAM2-UNet 5.43
Swin-Unet 6.57
WSRFNet 8.14
DeepLabv3 8.14
RETFound+UPerNet 9.86
M2MRF 10.71
17.57 FCT
15.86 SAM-Adapter
14.71 HACDR-Net
14.29 MedSAM
13.71 PMCNet
12.57 SAMed
12.14 CLC-Net
12.00 TC-Net
11.71 U-Net
Friedman χ² = 75.9, P < 0.001; Nemenyi critical difference at α = 0.05; n = 7 test sets
f
simulated
Published
Retrained
HACDR-Net
M2MRF
HRNet-W48
Swin-Unet
DeepLabv3
FCT
U-Net
SAMed
SAM-Adapter
MedSAM (full FT)
HRNet-W48
Swin-Unet
M2MRF
U-Net
HACDR-Net
DeepLabv3
SAMed
MedSAM (full FT)
SAM-Adapter
FCT
Rank on IDRiD (1 = best) · Kendall τ = 0.47, P = 0.07, n = 10